

Frequency

A frequency counts how often a value or category occurs.

In 0, 1, 1, 2, 2, 2, 3, 4, what is the frequency of 2?

Frequency tables

Number of downloads for 12 sessions:

0, 1, 2, 1, 3, 2, 0, 2, 1, 4, 2, 2

What frequency table summarizes these data?

Relative frequency

Relative frequency = frequency / total number of observations.

If 5 of 12 sessions have two downloads, what is their relative frequency?

Percentages

Seven of 20 printers are in use.

What percentage of the printers is in use?

Cumulative relative frequency

Cumulative relative frequency adds relative frequencies through a value.

If frequencies for 0, 1, 2, 3 downloads are 2, 3, 4, 1, what fraction have at most 2?

At most or less than?

Downloads: 0, 1, 1, 2, 2, 2, 3, 4

What percentage have fewer than 2 downloads?

More than a cutoff

Downloads: 0, 1, 1, 2, 2, 2, 3, 4

What percentage have more than 2 downloads?

Checking a table

A frequency table for 20 observations lists counts 4, 7, and 8.

What should we check before interpreting it?

Grouped data

Download times in seconds: 2, 4, 5, 7, 8, 9, 11, 14

**How many times belong in each interval:
0 to less than 5, 5 to less than 10, 10 to less than 15?**

Rounding

The mean of 4, 5, and 8 is required to one decimal place.

What answer should we report after calculating with full precision?

Rounding totals

Three equally frequent categories each have relative frequency $1/3$.

Why might displayed percentages fail to total exactly 100%?

Nominal level

Nominal data are categories with no inherent order.

What is the level of measurement for a printer's brand?

Ordinal level

Ordinal data have an order without defined numerical gaps.

What can “poor, fair, good, excellent” tell us about service?

Interval level

Interval data have meaningful differences but no meaningful absolute zero.

Does 20 degrees Celsius represent twice the temperature of 10 degrees Celsius?

Ratio level

Ratio data have meaningful differences and a zero that represents none of the measured quantity.

Does a 20-minute download take twice as long as a 10-minute download?

Measurement level

A tournament records first, second, and third place.

What level of measurement is this?

Measurement level

A file list records sizes in bytes.

What level of measurement is this?

Choosing a summary

A survey codes favorite browsers as 1, 2, 3, and 4.

Would the mean of these codes answer a useful question?